

Results in Harmonic Analysis

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Abstract

This write-up explores some classical results of Harmonic Analysis. We begin with some small and easy to prove results such as the Loomis Whitney Inequality, Isoperimetric Inequality etc. and then go on to prove bigger theorems such as the Hardy-Littlewood-Sobolev Inequality. After this, we move on to our main topic for this write up which explores some properties of oscillatory integrals under certain geometric constraints.

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1. Introduction

Harmonic Analysis is a field of Mathematics that uses methods of Real Analysis, Differential Equations and Geometry to study the representation and transformation of functions. This write-up in particular begins with a quick introduction to some geometric results before we jump into some analytic results where we try to look at $\mathcal{L}^p$ estimates for different operators.

In section 2, we study geometric inequalities like the Loomis–Whitney and Isoperimetric inequalities, which serve as prerequisite results to more analytic results such as the Hardy-Littlewood-Sobolev Inequality. This section serves the purpose of telling us how different geometric constraints on a function directly affect its $\mathcal{L}^p$ norms and derivative estimates.

Section 3 begins with a recollection of results like the Vitali Covering Lemma and Ball Doubling Conjecture which builds the stage for the Tube Doubling Conjecture in Section 4. In this section, we also look at the convolution operator with kernel $|x|^{-\alpha}$ and its properties. This helps us in building our intuition for the latter sections where the kernel of the operator includes an oscillating part.

In section 4, we simply state the Tube Doubling Conjecture and describe briefly the construction of the Besicovitch set which helps us set the stage for our main result proved in section 5. Finally, in section 5 we first explore oscillatory integrals and look at some examples of functions to get an idea of the behaviour of these integrals. We then provide a proof of the statement:

For $\alpha = \frac{n+1}{2}$, the conjecture $\|\tilde{T}_\alpha f\|_p \lesssim \|f\|_p$ is false for all $p \neq 2$ where $\tilde{T}_\alpha$ is the convolution operator with kernel $[1 + |x|]^{-\alpha} \cos|x|$

This result, due to Fefferman, is a major counterexample in Harmonic Analysis. It demonstrates that even though certain convolution operators with positive kernels obey $\mathcal{L}^p$ bounds under suitable conditions, introducing an oscillating kernel can completely break these estimates. The proof of this statement relies on the geometry of the construction of tubes as mentioned in the previous section and on the oscillatory behavior of the function we define.

2. The Sobolev Inequality

Let X be a set of unit cubes in the unit cubical lattice in $\mathbf{R}^n$. We denote by $|X|$ the volume of X .

Let

$$\begin{aligned} \pi_j : \mathbf{R}^n &\rightarrow \mathbf{R}^{n-1} \\ (x_1, \dots, x_j, \dots, x_n) &\mapsto (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \end{aligned}$$

i.e it is the map which removes the j^{th} coordinate. We will denote by $\hat{x}_j$ the point which removes the x_j component.

This map is clearly a continuous and smooth map.

Theorem 2.1 If $|\pi_j(X)| \leq A$, then $|X| \lesssim A^{\frac{n}{n-1}}$

We say that a column is a set of cubes obtained by starting at any cube and taking all the cubes along a line in the x_j direction.

Lemma 2.2 If $|\pi_j(X)| \leq B$ for all j , then there exists a column of cubes with the number of cubes being between 1 and $B^{\frac{1}{n-1}}$

Proof of Lemma 1.2

Suppose by contradiction that this is not true, i.e, each column has greater than $B^{\frac{1}{n-1}}$ cubes. Then this means that there are greater than $B^{\frac{1}{n-1}}$ cubes of X on some line, say A_1 parallel to the x_1 axis.

If $p \in A_1$ and lies in a cube of X , then the line through p parallel to the x_2 axis must intersect more than $B^{\frac{1}{n-1}}$ cubes of X .

Let A_2 be the two plane containing A_1 and parallel to the (x_1, x_2) axis.

Now this must intersect $B^{\frac{2}{n-1}}$ cubes of X .

Continuing in this way, we get $(n-1)$ plane A_{n-1} which is parallel to the $(x_1, \dots, x_{n-1})$ plane and intersecting more than B cubes of X .

But this implies that $|\pi_n(X)| > B$ which is a contradiction.

QED

Proof of Theorem 1.1: This is clear by induction on the lemma.

We can now generalize this theorem as:

Theorem 2.3 Generalized Loomis Whitney: If U is an open set in $\mathbf{R}^n$ with $|\pi_j(U)| \leq A$, then

$$|U| \lesssim A^{\frac{n}{n-1}}$$

Proof:

Take $U_\epsilon \subset U$ to be the union of ϵ cubes in the ϵ lattice.

Then,

$$|U_\epsilon| \leq A^{\frac{n}{n-1}}$$

and

$$|U_\epsilon| \rightarrow |U|$$

QED

Corollary 2.4 Isoperimetric Inequality: If U is a bounded opne set in $\mathbf{R}^n$, then

$$Vol_n(U) \lesssim Vol_{n-1}(\partial U)^{\frac{n}{n-1}}$$

Proof:

Since U is bounded, any line that intersects U must also intersect ∂U and so

$$\pi_j(U) \subset \pi_j(\partial U)$$

As π_j can only decrease $(n-1)$ dimensional volumes, thus we have that

$$|\pi_j(U)| \leq Vol_{n-1}(\partial U)$$

By Theorem 2.3,

$$|U| \lesssim (\max_j |\pi_j(U)|)^{\frac{n}{n-1}} \leq Vol_{n-1}(\partial U)^{\frac{n}{n-1}}$$

QED

Let $u \in C_0^1(\mathbf{R}^n)$ such that $\int |\nabla u| = 1$ and $h \in \mathbf{R}^n$. Then we define:

$$S_u(h) := \{x \in \mathbf{R}^n : |u(x)| > h\}$$

Lemma 2.5 If $u \in C_0^1(\mathbf{R}^n)$, then for all j ,

$$|\pi_j(S_u(h))| \leq h^{-1} \|\nabla u\|_{\mathcal{L}^1}$$

Proof: For $x \in S_u(h)$, take a line l in the x_j direction. Since u is compactly supported, it eventually reaches a point x' where $u(x') = 0$.

By the Fundamental Theorem of Calculus,

$$h < |u(x) - u(x')| \leq \int_l |\nabla u|$$

Thus we get

$$\|\nabla u\|_{\mathcal{L}^1} = \int |\nabla u| \geq \int_{\pi_j(S_u(h) \times \mathbf{R})} |\nabla u| = \int_{\pi_j(S_u(h))} \left(\int_{\mathbf{R}} |\nabla u| dx_j \right) d\hat{x}_j \geq |\pi_j(S_u(h))| \cdot h$$

QED

Now combining this lemma with Theorem 2.3 we get

$$|S_u(h)| \lesssim h^{\frac{-n}{n-1}} \cdot \|\nabla u\|_{\mathcal{L}^1}^{\frac{n}{n-1}}$$

We are now close to our main goal in this section, which is to prove the Sobolev Inequality.

Now for each k , define

$$T_u(k) := \{x \in \mathbf{R}^n : 2^k < |u(x)| \leq 2^{k+1}\}$$

Then for any p ,

$$\|u\|_{\mathcal{L}^p}^p = \int |u|^p \sim \sum_k |T_u(k)|^{2kp}$$

and since by definition, $|T_u(k)| \leq |S_u(2^k)|$, we get

$$|T_u(k)| 2^{\frac{kn}{n-1}} \lesssim \|\nabla u\|_{\mathcal{L}^1}^{\frac{n}{n-1}}$$

Lemma 2.6 If $u \in C_0^1(\mathbf{R}^n)$, then for any j ,

$$|\pi_j(T_u(k))| \lesssim 2^{-k} \int_{T_u(k-1)} |\nabla u|$$

Proof:

Suppose $x \in T_u(k)$ i.e $2^k < |u(x)| \leq 2^{k+1}$ and consider a line l_x through x in the x_j direction. Since u is compactly supported, there exists x' such that $u(x') = 0$.

Between x and x' , there exists a region of l_x where $2^{k-1} \leq |u| \leq 2^k$ and by definition, this region is $T_u(k-1)$. In this region, there must be at least one line segment where $|u| = 2^k$ and $|u| = 2^{k-1}$ at the end points. Then by the Fundamental Theorem of Calculus,

$$\int_{l_x \cap T_u(k-1)} |\nabla u| \geq 2^{k-1}$$

So l_x only depends on the choice of coordinates apart from the x_j coordinate. Thus we get,

$$\int_{T_u(k-1)} |\nabla u| \geq \int_{\pi_j(T_u(k))} \left(\int_{l_x \cap T_u(k-1)} |\nabla u| dx_j \right) d\hat{x}_j \geq 2^{k-1} |\pi_j(T_u(k))|$$

QED

Theorem 2.7 Sobolev's Inequality: If $u \in C_0^1(\mathbf{R}^n)$, then

$$\|u\|_{\mathcal{L}^{\frac{n}{n-1}}} \lesssim \|\nabla u\|_{\mathcal{L}^1}$$

Proof:

Combining Theorem 2.3 and the above lemma, we get

$$|T_u(k)| \lesssim 2^{\frac{-kn}{n-1}} \left(\int_{T_u(k-1)} |\nabla u| \right)^{\frac{n}{n-1}}$$

So if we sum over the different values of k , we get

$$\int |u|^{\frac{n}{n-1}} \sim \sum_{k=-\infty}^{\infty} |T_u(k)| \cdot 2^{\frac{kn}{n-1}} \lesssim \sum_k \left(\int_{T_u(k-1)} |\nabla u| \right)^{\frac{n}{n-1}} \leq \left(\int_{\mathbf{R}^n} |\nabla u| \right)^{\frac{n}{n-1}}$$

i.e.

$$\|u\|_{\mathcal{L}^{\frac{n}{n-1}}} \leq \|\nabla u\|_{\mathcal{L}^1}$$

QED

Remark: This statement is infact only true for $p = \frac{n}{n-1}$ and fails to hold in other $\mathcal{L}^p$ spaces.

3. The Hardy-Littlewood-Sobolev Inequality

Let us define an operator T which maps functions of $\mathbf{R}^n$ to functions of $\mathbf{R}^n$. Our aim is to find $\mathcal{L}^p$ estimates such that

$$\|Tf\|_{\mathcal{L}^q} \leq C\|f\|_{\mathcal{L}^p}$$

The Hardy-Littlewood-Sobolev Inequality is a similar statement that gives us estimates for an important operator that we define below.

Let $0 < \alpha < n$ and define $T_\alpha f$ as the convolution operator given by

$$T_\alpha f := f * |x|^{-\alpha}$$

i.e

$$T_\alpha f(x) = \int f(y) |x - y|^{-\alpha} dy$$

Theorem 3.1 Hardy-Littlewood-Sobolev Inequality: If $p > 1$ and $\alpha = n(1 - \frac{1}{q} + \frac{1}{p})$, then

$$\|T_\alpha f\|_q \leq C\|f\|_p$$

where C depends on n, p and q .

Before we begin with the proof of this very important inequality, we will briefly go over some short but very relevant results that may have been done in courses of undergraduate analysis.

3.1. Important Results

Lemma 3.2 Vitali Covering: If $\{B_i\}_{i \in I}$ is a finite collection of balls in $\mathbf{R}^n$, then there exists a subset $J \subset I$ such that $\{B_j\}_{j \in J}$ are disjoint but $\bigcup_i B_i \subset \bigcup_j 3B_j$.

We will skip the proof and state a slightly more important result:

Theorem 3.3 Ball Doubling Conjecture: If $\{B_i\}_{i \in I}$ is a finite collection of balls, then

$$|\bigcup 2B_i| \leq 6^n |\bigcup B_i|$$

Proof:

We apply Vitali to the set of balls $\{2B_i\}_{i \in I}$.

There exists a subset $J \subset I$ such that $\{2B_j\}_{j \in J}$ are disjoint but

$$\bigcup_i 2B_i \subset \bigcup_j 6B_j$$

In particular, $\{B_j\}$ are disjoint and so

$$|\bigcup_i 2B_i| \leq |\bigcup_j 6B_j| \leq 6^n \sum_j |B_j| = 6^n |\bigcup_j B_j|$$

QED

Lemma 3.4 Vitali for infinite case: Suppose $\{B_i\}_{i \in I}$ is a collection of balls in $\mathbf{R}^n$ and suppose that there exists a constant M such that any disjoint subset of balls $\{B_i\}$ has total volume at most M . Then, there exists a subset $J \subset I$ such that $\{B_j\}_{j \in J}$ are disjoint but $\bigcup_i B_i \subset \bigcup_j 4B_j$.

Since the proof is similar to the proof of Lemma 3.2, we will skip it.

3.2. Useful Inequalities

We introduce a new concept, known as the Hardy Littlewood Maximal function and define it as follows: For a measurable subset $A \in \mathbf{R}^n$, if

$$\oint_A f := \frac{1}{Vol A} \int_A f$$

we define the Hardy Littlewood Maximal function as

$$Mf(x) := \sup_r \oint_{B(x,r)} |f|$$

We remark that if f is continuous, then $Mf(x) \geq |f(x)|$. Further, the Hardy Littlewood Maximal function is lower semi continuous whenever $f \in \mathcal{L}_{loc}^1$. Indeed,

suppose $\epsilon > 0$. By the definition of supremum, there exists some $r_0 > 0$ such that

$$\frac{1}{|Vol(B(x_0, r_0))|} \int_{B(x_0, r_0)} |f(y)| dy > Mf(x_0) - \epsilon$$

Now, the function

$$x \mapsto \frac{1}{|Vol(B(x, r_0))|} \int_{B(x, r_0)} |f(y)| dy$$

is continuous. So for x sufficiently close to x_0 , we have that

$$\frac{1}{|Vol(B(x, r_0))|} \int_{B(x, r_0)} |f(y)| dy > \frac{1}{|Vol(B(x_0, r_0))|} \int_{B(x_0, r_0)} |f(y)| dy - \epsilon > Mf(x_0) - 2\epsilon$$

Since

$$Mf(x) \geq \frac{1}{|Vol(B(x, r_0))|} \int_{B(x, r_0)} |f(y)| dy \quad \forall x$$

Thus, we get that

$$Mf(x) \geq Mf(x_0) - 2\epsilon$$

and so

$$\liminf_{x \rightarrow x_0} Mf(x) \geq Mf(x_0)$$

QED

Lemma 3.5 For each $h > 0$, $|S_{Mf}(h)| \lesssim h^{-1} \|f\|_1$

Proof:

By definition of supremum, for each $x \in S_{Mf}(h)$, there exists $r(x)$ such that

$$\begin{aligned} \oint_{B(x, r(x))} |f| &\geq h \\ \implies \int_{B(x, r(x))} |f| &\geq h |B(x, r(x))| \end{aligned}$$

These balls, say $\mathcal{B} := \{B(x, r(x)) : x \in S_{Mf}(h)\}$ cover $S_{Mf}(h)$ and so we can apply Vitali. Suppose $\{B_\alpha\}$ is a disjoint sub-collection of these balls. Then

$$\int_{\mathbf{R}^n} |f| \geq \int_{\bigcup_\alpha B_\alpha} |f| = \sum_\alpha \int_{B_\alpha} |f| \geq h \sum_\alpha |B_\alpha|$$

and so

$$\sum_\alpha |B_\alpha| \leq h^{-1} \|f\|_1$$

Thus we have a uniform bound and through application of Vitali for the infinite case 3.4, we know that there exists disjoint $\{B_j\}$ such that

$$S_{Mf}(h) \subset \bigcup_j 4B_j$$

and hence

$$|S_{Mf}(h)| \lesssim \sum_j |B_j| \leq h^{-1} \int_{\bigcup B_j} |f| \leq h^{-1} \|f\|_1$$

which concludes the proof.

As with earlier notations, we see that

$$T_{Mf}(k) \subset S_{Mf}(2^k)$$

and

$$\int |Mf|^p \sim \sum_{k=-\infty}^{\infty} |T_{Mf}(k)| 2^{kp} \leq \sum_k |S_{Mf}(2^k)| 2^{kp}$$

Then the previous lemma further gives us that

$$\int |Mf|^p \lesssim \sum_k 2^{-k} 2^{kp} \|f\|_1$$

which is a divergent sum. So we need to make a slight modification of the previous lemma.

Lemma 3.6

$$|S_{Mf}(h)| \lesssim h^{-1} \int_{S_f(\frac{h}{2})} |f|$$

Proof:

In the proof of the previous lemma, we found disjoint balls $\{B_j\}$ such that $4B_j$ covers $S_{Mf}(h)$ and such that for each j , we get

$$\int_{B_j} |f| \geq h|B_j|$$

Now,

$$\int_{B_j \setminus S_f(\frac{h}{2})} |f| \leq \frac{h}{2}|B_j| \implies \int_{B_j \cap S_f(\frac{h}{2})} |f| \geq \frac{h}{2}|B_j|$$

Thus we get

$$\begin{aligned} |S_{Mf}(h)| &\leq |\bigcup_j 4B_j| \lesssim \sum_j |B_j| \lesssim h^{-1} \sum_j \int_{B_j \cap S_f(\frac{h}{2})} |f| \\ &\leq h^{-1} \int_{S_f(\frac{h}{2})} |f| \end{aligned}$$

with the final inequality coming from the fact that B_j being disjoint implies that $B_j \cap S_f(\frac{h}{2})$ are disjoint and $B_j \cap S_f(\frac{h}{2})$ is contained in $S_f(\frac{h}{2})$.

QED

We are now getting closer to our goal of this section, which is proving the Hardy-Littlewood-Sobolev Inequality.

Theorem 3.7 For any dimension n and $p > 1$, there exists a constant C depending on n and p such that

$$\|Mf\|_{\mathcal{L}^p} \leq C\|f\|_{\mathcal{L}^p}$$

Proof:

As seen earlier,

$$\int |Mf|^p \lesssim \sum_{k=-\infty}^{\infty} |S_{Mf}(2^k)| 2^{kp}$$

Now using the lemma we just prove, we get

$$\begin{aligned} \sum_{k=-\infty}^{\infty} |S_{Mf}(2^k)| 2^{kp} &\leq \sum_k 2^{k(p-1)} \int_{S_f(2^{k-1})} |f| \\ &= \int_{\mathbf{R}^n} |f| \left(\sum_{k: 2^{k-1} \leq |f|} 2^{k(p-1)} \right) \end{aligned}$$

Since $p > 1$, this sum becomes a geometric sum and the largest term dominates. This happens when $2^k \sim |f|$ and has size $\sim |f|^{p-1}$. So the above equation is

$$\sim \int |f| |f|^{p-1} = \int_{\mathbf{R}^n} |f|^p$$

and so

$$\|Mf\|_p \lesssim \|f\|_p$$

QED

3.3. Proof of The Hardy-Littlewood-Sobolev Inequality

We now arrive at the heart of this section. We will begin with a lemma before we jump into the proof.

Lemma 3.8

$$T_\alpha f(x) = \int_0^\infty r^{n-\alpha-1} \left(\oint_{B(x,r)} f \right) dr$$

The proof of this is completely computational, so we will quickly summarize the proof without going too much into the details of the calculations

Proof:

By definition,

$$T_\alpha f(x) = \int f(y) |x - y|^{-\alpha} dy$$

So let $z = y - x$. If we use the fact that Lebesgue measure is invariant under translation and then switch to polar coordinates, we will get

$$T_\alpha f(x) = \int_0^\infty \int_{S^{n-1}} f(x + r\theta) r^{n-1-\alpha} d\sigma(\theta) dr$$

Again, by definition,

$$\int_0^\infty r^{n-\alpha-1} \left(\oint_{B(x,r)} f \right) dr = \int_0^\infty r^{n-\alpha-1} \left(\frac{1}{|B(x,r)|} \int_{B(x,r)} f(y) dy \right) dr$$

Now letting

$$|B(x,r)| = \omega_n r^n$$

where ω_n is the volume of the unit ball in $\mathbf{R}^n$, we can simplify our computations and arrive at the required conclusion.

Proof of Theorem 3.1

We want to first find an upper bound for $\oint_{B(x,r)} f$ but that is easy as from the definition of $Mf(x)$, we get that

$$\left| \oint_{B(x,r)} f \right| \leq Mf(x) \tag{1}$$

Also, we can apply Hölder's inequality to get

$$\left| \oint_{B(x,r)} f \right| \lesssim r^{-n} \int_{B(x,r)} |f| \lesssim r^{-n} \|f\|_p r^{\frac{n(p-1)}{p}} = r^{\frac{-n}{p}} \|f\|_p \tag{2}$$

Combining both the equations 1 and 2 gives us,

$$\left| \oint_{B(x,r)} f \right| \leq \min(Mf(x), r^{\frac{-n}{p}} \|f\|_p)$$

Plugging this into the above lemma we get,

$$T_\alpha f(x) \lesssim \int_0^\infty r^{n-\alpha-1} \min(Mf(x), r^{\frac{-n}{p}} \|f\|_p) dr$$

We must now compute this. Define $r_0 = \left(\frac{\|f\|_p}{Mf(x)} \right)^{\frac{p}{n}}$

So when $r < r_0$ then the minimum is $Mf(x)$ and when $r > r_0$ then the minimum is $r^{\frac{-n}{p}} \|f\|_p$.

So we can divide the integral as follows:

$$\int_0^\infty r^{n-\alpha-1} \min(Mf(x), r^{\frac{-n}{p}} \|f\|_p) dr = \int_0^{r_0} r^{n-\alpha-1} Mf(x) dr + \int_{r_0}^\infty r^{n-\alpha-1} r^{\frac{-n}{p}} \|f\|_p dr$$

Again, we are skipping the details of the computation but the first integral on the right turns out to be equal to

$$\frac{1}{n-\alpha} Mf(x)^{1-\frac{p(n-\alpha)}{n}} \|f\|_p^{\frac{p(n-\alpha)}{n}}$$

and for the second integral, it turns out that it converges if and only if

$$\alpha > n(1 - \frac{1}{p})$$

and then computation shows that the integral is equal to

$$\frac{1}{\alpha - n(1 - \frac{1}{p})} Mf(x)^{\frac{-p(n(1-\frac{1}{p})-\alpha)}{n}} \|f\|_p^{1+\frac{p(n(1-\frac{1}{p})-\alpha)}{n}}$$

Combining both the integrals, we get the final integral in the form

$$|T_\alpha f(x)| \lesssim Mf(x)^A \|f\|_p^B$$

where $A = 1 - \frac{p(n-\alpha)}{n}$ and $B = \frac{p(n-\alpha)}{n}$ which also gives us that $A + B = 1$.

Finally, we want to bound $\|T_\alpha f\|^q$ for some q . From above, we already have

$$\int |T_\alpha f(x)|^q \lesssim \|f\|_p^{Bq} \int |Mf(x)|^{Aq} \quad (3)$$

If we choose q such that $Aq = p$, equation (3) gives is

$$\|T_\alpha f\|_{\mathcal{L}^q}^q \lesssim \|f\|_p^{Bq} \cdot \|Mf\|_p^{Aq}$$

So finally, by Theorem 3.7, we get that if $p > 1$, we have

$$\|T_\alpha f\|_{\mathcal{L}^q}^q \lesssim \|f\|_p^{Bq} \cdot \|Mf\|_p^{Aq} \lesssim \|f\|_{\mathcal{L}^p}^{(A+B)q} = \|f\|_{\mathcal{L}^p}^q$$

and so

$$\|T_\alpha f\|_{\mathcal{L}^q} \lesssim \|f\|_{\mathcal{L}^p}$$

if $p > 1$ and $Aq = p$. From above computations, we can also conclude that $\alpha = n(1 - \frac{1}{q} + \frac{1}{p})$.

QED

4. The Tube Doubling Conjecture and Besicovitch's Construction

In this section, we will very briefly discuss the the Tube Doubling Conjecture, which is analogous to the Ball Doubling Conjecture that we discussed earlier in the previous section. We will then proceed to give a construction of a Besicovitch set which acts as a counter example to the conjecture. We will omit the details of the proofs in this section and will assume they are true for the details of the proofs are not relevant to our main goal in this write up. One could refer to [1] for the details.

The Problem: Suppose $T_i \subset \mathbf{R}^n$ are cylindrical tubes of radius 1 and length N and suppose that $2T_i$ are the concentric tubes obtained after dilating the T_i around the center by a

factor of 2. The problem asks whether there exists a C_n such that for any set of $1 \times N$ tubes T_i in $\mathbf{R}^n$,

$$|\bigcup_i 2T_i| \leq C_n |\bigcup_i T_i|$$

The answer to this question, surprisingly, turns out to be no and a counterexample was given by Besicovitch in which

$$|\bigcup_i 2T_i| \sim \frac{\log N}{\log \log N} |\bigcup_i T_i|$$

This was later refined by Schoenberg where the $\log \log N$ factor was removed. Formally, the conjecture is:

The Tube Doubling Conjecture: For any dimension n and for any $\epsilon > 0$, there exists $C_n(\epsilon)$ so that if T_i are tubes of radius 1 and length N , then

$$|\bigcup_i 2T_i| \leq C_n(\epsilon) N^\epsilon |\bigcup_i T_i|$$

This conjecture was already solved for dimension 2 but has very recently also been solved for dimension 3 by Hong Wang and Joshua Zahl. There is another famous conjecture, known as the Kakeya conjecture which is closely related to the Tube Doubling Conjecture. The Tube Doubling Conjecture implies bounds that are consistent with Kakeya-type estimates. We describe the tube version of the Kakeya Conjecture below.

Definition: For a tube T , we write $v(T) \in S^{n-1}$ for the unit vector parallel to the axis of symmetry of T . We will call $v(T)$ as the direction of the tube T . Naturally, we have 2 possible choices with different signs.

Definition: Suppose $T_i \subset \mathbf{R}^n$ are tubes of radius 1 and length N . We say that $\{T_i\}$ is a Kakeya set of tubes if $v\{T_i\}$ is $\frac{1}{N}$ separated and $\frac{2}{N}$ dense in S^{n-1} i.e for any $i \neq j$,

$$\|v(T_i) - v(T_j)\| \geq \frac{1}{N}$$

and

$$\forall x \in S^{n-1}, \exists i : \|v(T_i) - x\| \leq \frac{2}{N}$$

Kakeya's Conjecture: In any dimension $n \geq 2$, for every $\epsilon > 0$, there exists $C_{n,\epsilon}$ such that for any N and any Kakeya set of tubes T_i of dimension $1 \times N$,

$$|\bigcup_i T_i| \geq C_{n,\epsilon} N^{n-\epsilon}$$

We now give an example of a Besicovitch set. We will omit the details of the proof for this.

For integers $1 \leq j \leq N$, let $l_j : [0, 1] \rightarrow \mathbf{R}$ be a list of affine linear functions of the form

$$l_j(x) = \frac{j}{N}x + H(j)$$

We then let R_j be the $\frac{1}{N}$ neighborhoods of the graph of l_j which contain a rectangle of width $\frac{1}{N}$ and length 1.

The idea behind this construction is that these lines have varying slopes and the rectangles have a lot of overlap which helps it to spread direction across a wide range while keeping the area relatively small.

Our next job is to select $H(j)$ in such a way that the following theorem holds true.

Theorem 4.1 Suppose N is an integer of the form A^A for some very large integer A . If we choose the $H(j)$ correctly, then we have

$$|\bigcup_i R_j| \lesssim A^{-1} \lesssim \frac{\log N}{\log \log N}$$

The proof of this theorem begins with expanding $\frac{j}{N}$ in base A to get

$$\frac{j}{N} = \sum_{a=1}^A j(a) A^{-a}$$

where $j(a)$ are the digits in the base A decimal expansion of $\frac{j}{N}$. The different values of a represent different scales. We then choose our $H(j)$ such that the following proposition holds.

Proposition 4.2 Suppose $1 \leq b \leq A$. If $j(a) = j'(a)$ for $1 \leq a \leq b-1$, then for all $x \in [\frac{A-b}{A}, \frac{A-b+1}{A}]$,

$$|l_j(x) - l_{j'}(x)| \leq 4A^{-b}$$

Finally, we choose our $H(j)$ as

$$H(j) = \sum_{a=1}^A h(a) j(a) A^{-a}$$

where $h(a) \in [-1, 1]$. In fact, we can take $H(a) = -\frac{A-a}{A}$.

Denote by R_j^+ as the translation of R_j by $10v(R_j)$ and choose the direction of $v(R_j)$ with the x component being negative. As a corollary to the above theorem, we get

Corollary 4.3 Suppose N is an integer of the form A^A for some very large integer A . If we choose the $H(j)$ correctly, then we have

$$|\bigcup_i R_j| \lesssim \frac{\log \log N}{\log N} \sum_j |R_j|$$

and yet the R_j^+ are all disjoint.

5. Oscillatory Integrals and Fefferman's Counter Example

5.1. Oscillatory Integrals

In section 2, we had defined an operator T which was the convolution of a function f with $|x|^{-\alpha}$, specifically,

$$T_\alpha f = f * |x|^{-\alpha}$$

Let us call this kernel as K_α and we remark that this is always positive. In this section, we will look at a variation of this which makes the kernel into an oscillatory type kernel, which means it is sometimes negative and sometimes positive. Formally, let us define

$$\tilde{K}_\alpha := [1 + |x|]^{-\alpha} \cos|x|$$

and define our operator $\tilde{T}_\alpha$ as

$$\tilde{T}_\alpha f := f * \tilde{K}_\alpha$$

Notice that as $x \rightarrow 0$, $|\tilde{K}_\alpha(x)| \leq M$ for some $M > 0$ and when $|x| \geq 1$, we have

$$\begin{aligned} |\tilde{K}_\alpha(x)| &= |[1 + |x|]^{-\alpha} \cos|x|| \\ &\leq |[1 + |x|]^{-\alpha}| \\ &\leq C|x|^{-\alpha} \\ &\sim K_\alpha(x) \end{aligned}$$

Let us now consider an example. Suppose $f = \chi_{B_r}$ i.e f is the characteristic function on the ball of radius r centered at 0. Then,

$$\tilde{T}_\alpha f = \tilde{T}_\alpha \chi_{B_r} = \int_{B_r} \tilde{K}_\alpha(x - y) dy = \int_{B_r} [1 + |x - y|]^{-\alpha} \cos|x - y| dy$$

This is not an easy integral to compute. To look at a simpler case, we take $r = \frac{1}{100}$. Then at most points of x , the sign of $\cos|x - y|$ is constant for $y \in B_r$ and is ~ 1 . So at such a point x , we can say

$$|x - y| \sim |x|$$

and so a simple one line computation gives us

$$\tilde{T}_\alpha \chi_{B_r} \sim \int_{B_r} [1 + |x - y|]^{-\alpha} dy \sim |\tilde{K}_\alpha(x)|$$

So if we let $f_1 := \chi_{B_{\frac{1}{100}}}$, then for most x

$$|\tilde{T}_\alpha f_1(x)| \sim |\tilde{K}_\alpha(x)|$$

Also,

$$\|f_1\|_p = C_n r^n \sim 1$$

Here, by $\sim$ we mean that it is bounded upto a constant depending only on n and p . On the other hand,

$$\begin{aligned} \int |\tilde{T}_\alpha f_1|^p &= \int |[1 + |x - y|]^{-\alpha} \cos|x - y| \cdot \chi_{B_r}|^p dy \\ &= \int |\tilde{K}_\alpha(x)|^p dy \\ &\sim \int [1 + |x|]^{-\alpha p} dy \end{aligned}$$

In spherical coordinates, the integral becomes

$$\int r^{n-1-\alpha p} dr dS^{n-1}$$

which is finite if and only if

$$n - 1 - \alpha p < -1$$

i.e

$$\alpha p > n$$

Now we want to see what happens when $r \gg 1$. Let us focus at the point $x = 0$. So

$$\tilde{T}_\alpha f(0) = \int f(y) [1 + |y|]^{-\alpha} \cos|y| dy$$

We again face the same problem where we notice that due to multiple changes in the sign, lots of cancellations can take place and the integral is hard to compute. So we look at a particular f . Define

$$f_2 := \chi_{B_r} \text{sign}(\cos|y|)$$

Then all the contributions in the integral from f_2 becomes positive and we get

$$\|f\|_p = \left(\int_{B_r} |\text{sign}(\cos|y|)|^p \right)^{\frac{1}{p}} = |B_r|^{\frac{1}{p}} \sim r^{\frac{n}{p}} \quad (4)$$

Also,

$$\begin{aligned} |\tilde{T}_\alpha f_2(0)| &= \int_{B_r} [1 + |y|]^{-\alpha} \text{sign}(\cos|y|) dy \\ &= |S^{n-1}| \int_0^r [1 + \rho]^{-\alpha} |\cos \rho| \rho^{n-1} d\rho \\ &\sim |S^{n-1}| \int \rho^{n-1-\alpha} \cos|\rho| \quad \text{for } \rho \gg 1 \end{aligned}$$

Now as $|\cos \rho|$ is bounded and oscillates but is always non negative here, the above integral becomes

$$\sim \int_1^r \rho^{n-1-\alpha} \sim r^{n-\alpha}$$

In fact, it turns out that even for $|x| < \frac{1}{100}$,

$$|\tilde{T}_\alpha f_2(x)| \sim r^{n-\alpha}$$

A rigorous computation shows that

$$\|\tilde{T}_\alpha f_2\|_p \gtrsim r^{n-\alpha}$$

As seen in equation (4),

$$\|f_2\|_p \sim r^{\frac{n}{p}}$$

and so we get that

$$\|\tilde{T}_\alpha f_2\|_p \lesssim \|f_2\|_p \iff \frac{n}{p} \geq n - \alpha \quad (5)$$

Thus from the above example of f_1 and f_2 we get that

Proposition 5.1 If $\|\tilde{T}_\alpha f\|_p \leq \|f\|_p$ for the above f 's, then

$$\frac{n}{\alpha} \leq p \leq \frac{n}{n - \alpha}$$

We will now look at oscillating functions supported in a long thin tube.

Let T be a cylinder of length $L \gg 1$ and radius $\frac{\sqrt{L}}{1000}$ and let it be pointed at any direction. Let v_T be the unit vector perpendicular to the axis of the cylinder and define

$$f_T(x) := \chi_T(x) e^{i(v_T \cdot x)}$$

Let T^+ denote the cylinder we get by translating T by $10Lv_T$.

Proposition 5.2 Fix a dimension $n \geq 2$. For all L sufficiently large, the following holds.

If f_T and T^+ are defined as above, then for every $x \in T^+$, we have

$$|\tilde{T}_\alpha f_T(x)| \gtrsim L^{\frac{n+1}{2}-\alpha}$$

Corollary 5.3 If $\alpha < \frac{n+1}{2}$, then for every $p \in [1, \infty]$, as $L \rightarrow \infty$,

$$\frac{\|\tilde{T}_\alpha f_T\|_p}{\|f_T\|_p} \rightarrow \infty$$

which means that $\tilde{T}_\alpha$ does not obey $\|\tilde{T}_\alpha f\|_p \lesssim \|f\|_p$.

Proof:

Since T^+ is a translation of T , it has the same size as T . From the definition of f_T , it is clear that it has a size ~ 1 and is supported on T .

If $\alpha < \frac{n+1}{2}$, then $\tilde{T}_\alpha f_T$ has size $\gg 1$ on T^+ . Indeed, as support of f_T is T and $x \in T^+$, the integral in $\tilde{T}_\alpha f_T(x)$ looks at integration of kernel $\tilde{K}_\alpha(x-y)$ over $y \in T$ where $x-y \approx 10Lv_T$. Now,

$$\tilde{K}_\alpha(x-y) \sim [1 + |x-y|]^{-\alpha} \cos|x-y| \sim L^{-\alpha} \cos(L)$$

As the volume of $T \sim L^{\frac{n+1}{2}}$, this we get

$$\tilde{T}_\alpha f_T(x) \sim L^{\frac{n+1}{2}-\alpha}$$

and so we get

$$\|\tilde{T}_\alpha f_T\|_p \sim L^{\frac{n+1}{2}-\alpha} \|f_T\|_p$$

QED

Proof of Proposition 5.2

Let $x \in T^+$

$$\tilde{T}_\alpha f_T(x) = \int_T |x-y|^{-\alpha} \cos|x-y| e^{i(v_T \cdot y)} dy \quad (6)$$

We will now decompose the vector $x-y$ into a vector perpendicular to v_T and to a vector parallel to v_T .

Since the translation of T^+ is by a length of $10Lv_T$, by geometry of T and T^+ , we get that the parallel component has a length of at least $5L$ and the perpendicular component has a length of at most $\frac{\sqrt{L}}{1000}$. So we can apply the Pythagorus theorem to get

$$\begin{aligned} (v_T \cdot x - v_T \cdot y)^2 &\leq |x-y|^2 = (v_T \cdot x - v_T \cdot y)^2 + |(x-y)_\perp|^2 \\ &\leq (v_T \cdot x - v_T \cdot y)^2 + \frac{L}{10^6} \end{aligned}$$

Since $|v_T \cdot x - v_T \cdot y| \geq 5L$, we also get

$$||x-y| - |v_T \cdot x - v_T \cdot y|| \leq 10^{-6}$$

Now as $\cos(b) = \cos(-b)$ and $|\cos(a) - \cos(b)| \leq |a-b|$, we get

$$|\cos|x-y|| - \cos(v_T \cdot x - v_T \cdot y)| \leq 10^{-6}$$

Putting these estimates in equation (6), we get

$$\tilde{T}_\alpha f_T(x) = \int_T |x-y|^{-\alpha} \cos(v_T \cdot x - v_T \cdot y) e^{i(v_T \cdot y)} dy + \text{some small error}$$

where this small error is $\leq 10^{-6} \int_T |x-y|^{-\alpha} dy$

Now writing

$$\cos(a) = \frac{1}{2}(e^{ia} + e^{-ia})$$

we get

$$\tilde{T}_\alpha f_T(x) = \frac{1}{2} e^{i(v_T \cdot x)} \int_T |x - y|^{-\alpha} dy + \frac{1}{2} e^{-i(v_T \cdot x)} \int_T |x - y|^{-\alpha} e^{2i(v_T \cdot y)} dy + \text{some small error}$$

Now,

$$\int_T |x - y|^{-\alpha} dy \gtrsim L^{n-\alpha}$$

and

$$\int_T |x - y|^{-\alpha} e^{2i(v_T \cdot y)} dy \ll \int_T |x - y|^{-\alpha}$$

This is because $v_T \cdot y \in [0, L]$ along the cylinder and so many oscillations take place and since $|x - y|^{-\alpha}$ is slow varying, the integral averages out. So finally, we get that

$$|\tilde{T}_\alpha f_T(x)| \sim \int_T |x - y|^{-\alpha} dy$$

and as the volume of $T \sim L^{\frac{n+1}{2}}$ and $|x - y| \sim L$ for $y \in T$, we get that

$$|\tilde{T}_\alpha f_T(x)| \sim L^{\frac{n+1}{2}-\alpha} \quad (7)$$

QED

5.2. Fefferman's Counter Example

We now arrive at the main result of our write up. Till now, we have seen conditions for which

$$\|\tilde{T}_\alpha f\|_q \lesssim \|f\|_p$$

for different operators T . We are now going to look at the case where $\alpha = \frac{n+1}{2}$. Until 1971, it was believed that a similar $\mathcal{L}^p$ estimate would hold for the operator $\tilde{T}_\alpha$ which we defined above. But Charles Fefferman gave a counter example for this case by defining an oscillating function inside the tube constructed according to Besicovitch. In this section, we will look this function and read how Fefferman proved the claim false.

Theorem 5.4 $\|\tilde{T}_{\frac{n+1}{2}} f\|_p \lesssim \|f\|_p$ is false for all $p \neq 2$.

Proof:

We will first prove the case for $p > 2$. The case for $p < 2$ is similar and so we will skip it, but provide an idea at the end of this proof.

Let T_i be thin tubes of length L and radius $\frac{\sqrt{L}}{1000}$ arranged according to the Besicovitch construction and define

$$f_{T_i}(x) := \chi_{T_i}(x) e^{i(v_{T_i} \cdot x)}$$

and let

$$f := \sum_i f_{T_i}$$

Since $\tilde{T}_\alpha$ is linear, we get that

$$\tilde{T}_{\frac{n+1}{2}} f = \sum_i \tilde{T}_{\frac{n+1}{2}} f_{T_i}$$

Using Besicovitch's construction, we can arrange the T_i so that they are disjoint whereas the T_i^+ have a lot of overlap. Formally, we can generalize this as:

Theorem 5.5 Fix dimension $n \geq 2$. For any $L \geq 1$, there exists a finite set of disjoint tubes T_i with the property that

$$|\bigcup_i T_i^+| = \mu(L)^{-1} |\bigcup_i T_i|$$

where $\mu(L) \gtrsim \frac{\log L}{\log \log L}$

The proof of this is similar to the proof of Corollary 4.3 and we skip the details.

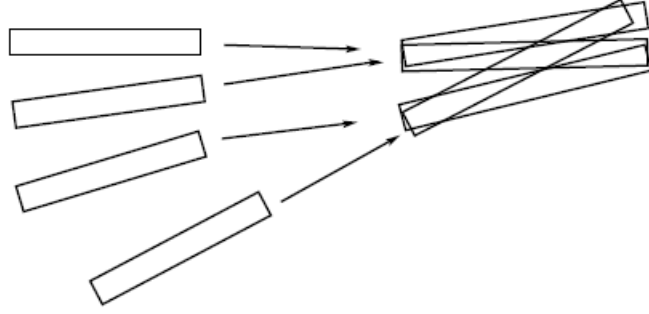


Figure 1: The disjoint tubes on the left are T_i and the overlapping ones on the right are the translated tubes T_i^+

Proof of Theorem 5.4 Continued:

Now, $|f_{T_i}(x)| = |\chi_{T_i}(x)|$ and since the tubes T_i are disjoint, we get

$$\|f\|_{\mathcal{L}_p}^p = \int |\sum_i f_{T_i}|^p = \sum \int |\chi_{T_i}(x)|^p dx = \sum_i |T_i|$$

We would like to look for a lower bound for $\|\tilde{T}_{\frac{n+1}{2}} f\|_p$. We define $f_{ran} := \sum_i \pm f_{T_i}$ where $\pm$ are selected at random. Regardless of the signs,

$$\|f_{ran}\|_p = \sum_i |T_i| \tag{8}$$

We will now require an important inequality given by Khintchin. The statement is as follows.

Proposition 5.6 If g_i 's are any functions and

$$g = \sum_i \pm g_i$$

with random signs, then for any $1 \leq p < \infty$, the average value of $\|g\|_p$ is

$$\sim_p \left\| \left(\sum_i |g_i|^2 \right)^{\frac{1}{2}} \right\|_p$$

which means there exists constants c_p and C_p such that

$$c_p \left\| \left(\sum_{i=1}^n |g_i|^2 \right)^{\frac{1}{2}} \right\|_p \leq \sum_{\epsilon \in \{\pm 1\}^n} 2^{-n} \left\| \sum_{i=1}^n \epsilon_i g_i \right\|_p \leq C_p \left\| \left(\sum_{i=1}^n |g_i|^2 \right)^{\frac{1}{2}} \right\|_p$$

In particular, we can apply 5.6 for $\tilde{T}_{\frac{n+1}{2}} f_{ran}$ to get that the average value of

$$\|\tilde{T}_{\frac{n+1}{2}} f_{ran}\|_p = \|\sum_i \pm \tilde{T}_{\frac{n+1}{2}} f_{T_i}\|_p$$

is

$$\sim \left(\int \left(\sum_i |\tilde{T}_{\frac{n+1}{2}} f_{T_i}|^2 \right)^{\frac{p}{2}} \right)^{\frac{1}{p}} = \left\| \left(\sum_i |\tilde{T}_{\frac{n+1}{2}} f_{T_i}|^2 \right)^{\frac{1}{2}} \right\|_p \quad (9)$$

Now from equation (7), we get that

$$|\tilde{T}_{\frac{n+1}{2}} f_{T_i}| \gtrsim 1$$

on T_i^+ and so

$$|\tilde{T}_{\frac{n+1}{2}} f_{T_i}|^2 \gtrsim 1 \quad \forall x \in T_i^+$$

which means it is bounded below by the number of indices i for which $x \in T_i^+$. This gives us

$$\sum_i |\tilde{T}_{\frac{n+1}{2}} f_{T_i}|^2 \gtrsim \sum_i \chi_{T_i^+}(x)$$

and so

$$\int \left(\sum_i |\tilde{T}_{\frac{n+1}{2}} f_{T_i}|^2 \right)^{\frac{p}{2}} \gtrsim \int \left(\sum_i \chi_{T_i^+} \right)^{\frac{p}{2}} \quad (10)$$

Now, by definition, the average value of $\left(\sum_i \chi_{T_i^+} \right)$ on $\bigcup_i T_i^+$ is $\mu(L)$. That means,

$$\mu(L) = \frac{1}{|\bigcup_i T_i^+|} \int \sum_i \chi_{T_i^+}(x) dx = \frac{\sum_i |T_i^+|}{|\bigcup_i T_i^+|}$$

Now we use a corollary of Hölder's inequality known as Jensen's inequality which states that for $f \geq 0$ measurable on a set $E \subset \mathbf{R}^n$ of finite measure and $p > 1$, we have

$$\int_E f(x)^p dx \geq |E|^{1-p} \left(\int_E f(x) dx \right)^p$$

Applying this inequality to the function

$$F(x) = \sum_i \chi_{T_i^+}(x)$$

Since

$$\text{supp}(F) = |\bigcup_i T_i^+| = \frac{1}{\mu(L)} \left| \sum_i T_i^+ \right|$$

and

$$\int F(x) dx = \left| \sum_i T_i^+ \right|$$

we get

$$\int \left(\sum_i \chi_{T_i^+}(x) \right)^{\frac{p}{2}} dx \gtrsim \mu(L)^{\frac{p}{2}-1} \left| \sum_i T_i^+ \right| \quad (11)$$

Now, combining equations (8), (9), (10) and (11), we get that

$$\begin{aligned} \text{average value of } \|\tilde{T}_{\frac{n+1}{2}} f_{ran}\|_p &\gtrsim \mu(L)^{\frac{p-2}{2p}} \sum_i |T_i^+| \\ &= \mu(L)^{\frac{p-2}{2p}} \|f_{ran}\|_p \end{aligned}$$

Now as $p > 2$, we have

$$\frac{p-2}{2p} > 0$$

and from the construction,

$$\mu(L) \rightarrow \infty$$

as $L \rightarrow \infty$ and so we finally arrive at the result:

$$\frac{\|\tilde{T}_{\frac{n+1}{2}} f_{ran}\|_p}{\|f_{ran}\|_p} \rightarrow \infty$$

as $\mu(L) \rightarrow \infty$.

QED

For the case $p < 2$, we take the same $f = f_{ran}$ but do the Besicovitch construction such that T_i overlap a lot and T_i^+ don't. A little check would show that

$$\|\tilde{T}_{\frac{n+1}{2}} f_{ran}\|_p^p \gtrsim \sum_i |T_i|$$

and that each point in $\bigcup_i T_i$ lies in $\mu(L)$ different tubes T_i . So one would expect

$$|f_{ran}(x)| \sim \mu(L)^{\frac{1}{2}}$$

for $x \in \bigcup_i T_i$ and hence we would get

$$\|f_{ran}\|_p^p \lesssim \mu(L)^{\frac{p}{2}} \left| \bigcup_i T_i \right| \sim \mu(L)^{\frac{p-2}{2}} \sum_i |T_i|$$

and since $p < 2$,

$$\mu(L)^{\frac{p-2}{2}} \rightarrow 0$$

which completes the proof. The arguments for this case can also be made rigorous using Khinchin's inequality.

6. Remark

This report was written as a part of the short term mandatory semester project for the M1 Mathematics curriculum at Université Sorbonne Paris Nord. Everything that we did in this report can be found in Larry Guth's book 'Polynomial Methods in Combinatorics' [1]. We have barely filled in the details whenever necessary.

References

- [1] Larry Guth. *Polynomial Methods in Combinatorics*, volume 64 of *University Lecture Series*. American Mathematical Society, Providence, RI, 2016. ISBN 978-1-4704-2890-7.